

Acaplamiento long-SSH.

$$\begin{aligned}\hat{H}_{int} &= (\hat{a}^\dagger - \hat{a}) \sum_{j,\sigma} i Y_j (b_{j\sigma}^\dagger - b_{j\sigma}) \\ &= \frac{(\hat{a}^\dagger - \hat{a})}{\sqrt{N}} \sum_{j,\sigma,k} Y_{j\sigma} (e^{ikj} b_{k\sigma}^\dagger - e^{ikj} b_{k\sigma}) \\ \hat{H}_{int} &= (\hat{a}^\dagger - \hat{a}) \sum_{k,\sigma} (Y_{k\sigma} b_{k\sigma}^\dagger - Y_{k\sigma}^* b_{k\sigma})\end{aligned}$$

$$Y_{k\sigma} = \frac{1}{\sqrt{N}} \sum_j Y_j e^{ikj}$$

• Acaplamiento constante:

Acaplamiento con A puede ser distinto al de B

$$Y_{k\sigma} = \frac{Y_\sigma}{\sqrt{N}} \sum_j e^{ikj} = \frac{Y_\sigma}{\sqrt{N}} S_{k\sigma} \cdot N$$

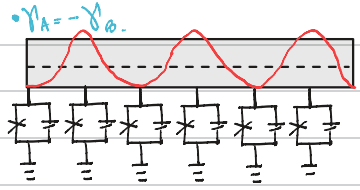
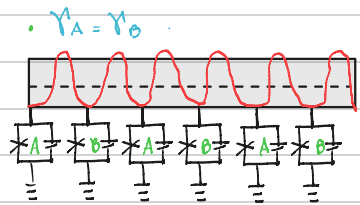
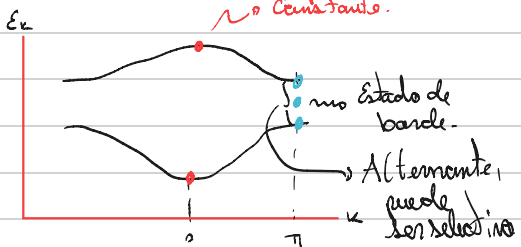
$$\hat{H}_{int} = \sqrt{N} \sum_\sigma Y_\sigma (b_{0\sigma}^\dagger - b_{0\sigma}) (\hat{a}^\dagger - \hat{a})$$

• Acaplamiento alternante.

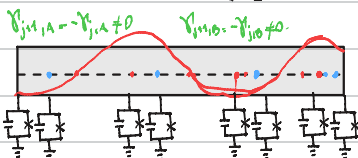
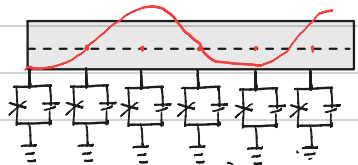
$$Y_{j+1,\sigma} = -Y_{j,\sigma} \rightarrow Y_{k\sigma} = \frac{Y_\sigma}{\sqrt{N}} N \delta_{k,\pi}$$

$$\hat{H}_{int} = \sqrt{N} \sum_\sigma Y_\sigma (b_{\pi\sigma}^\dagger - b_{\pi\sigma}) (\hat{a}^\dagger - \hat{a})$$

Representando en las bandas:



$$Y_{j+1,\sigma} = -Y_{j,\sigma} \neq 0, Y_{j\sigma} = 0, \forall j$$



Se puede hacer

$$Y_{j,0} = Y_{j,\pi} \text{ si los capacitores son distintos.}$$

En el caso general No puede ser una sola para todos los k .

Nuevamente tenemos un Hamiltoniano cuadrático que podemos diagonalizar por una transf. de Bogoliubov:

$$\hat{H} = \sum_k E_1 c_k^\dagger c_k + E_2 c_k^\dagger c_{-k} + E_3 c_k^\dagger c_{-k}^\dagger$$

Reemplazando

$$b_{k\sigma} = u_k^\sigma a_k + v_k^\sigma a_{-k}^\dagger + w_k^\sigma \beta_k + z_k^\sigma \beta_{-k}^\dagger$$

$$\sum_{k\sigma} (\gamma_{k\sigma} b_{k\sigma}^\dagger - \gamma_{k\sigma}^* b_{k\sigma}) = \sum_{k\sigma} \left\{ \gamma_{k\sigma}^* (u_k^\sigma - u_k^\sigma) a_k + \gamma_{k\sigma} (u_k^\sigma - v_k^\sigma) a_{-k}^\dagger + \gamma_{k\sigma}^* (z_k^\sigma - w_k^\sigma) \beta_k + \gamma_{k\sigma} (w_k^\sigma - z_k^\sigma) \beta_{-k}^\dagger \right\}$$

$\gamma_{kA} \rightarrow \gamma_{kA} e^{-i\phi_A}$
 Por el gauge en A

$$\tilde{H}_{int} = (\hat{a}^\dagger - \hat{a}) \sum_{k\sigma} \left\{ (u_k^\sigma - v_k^\sigma) [\gamma_{k\sigma} a_k^\dagger - \gamma_{k\sigma}^* a_k] + (w_k^\sigma - z_k^\sigma) [\gamma_{k\sigma} \beta_k^\dagger - \gamma_{k\sigma}^* \beta_k] \right\}$$

Definimos $\gamma_k = \sum_{\sigma} (u_k^\sigma - v_k^\sigma) \gamma_{k\sigma} \rightarrow \gamma_k^\sigma = \gamma_k, \sigma = A$
 $\gamma_k' = \sum_{\sigma} (w_k^\sigma - z_k^\sigma) \gamma_{k\sigma} \rightarrow \gamma_k' = \gamma_k', \sigma = B$

$$c_k^\sigma = \begin{cases} a_k, & \sigma = A \\ \beta_k, & \sigma = B \end{cases}$$

$$\tilde{H}_{int} = (\hat{a}^\dagger - \hat{a}) \sum_{k\sigma} (\gamma_k^\sigma c_k^\sigma - \gamma_k^{\sigma*} c_k^\sigma)$$

Por lo que el hamiltoniano completo es:

$$\hat{H} = \sum_{\vec{k}, \sigma} E_{\vec{k}, \sigma}^0 c_{\vec{k}, \sigma}^\dagger c_{\vec{k}, \sigma} + \hbar \omega_c (\hat{a}^\dagger \hat{a} + (\hat{a}^\dagger - \hat{a})) \sum_{\vec{k}, \sigma} (\eta_{\vec{k}, \sigma}^\dagger c_{\vec{k}, \sigma}^\dagger - \eta_{\vec{k}, \sigma}^* c_{\vec{k}, \sigma})$$

Definimos el espinor $\vec{d} = \begin{pmatrix} c_{\vec{k}, \sigma} \\ \hat{a} \\ c_{\vec{k}, \sigma}^\dagger \\ \hat{a}^\dagger \end{pmatrix}$, $\vec{C}_{\vec{k}, \sigma} = \begin{pmatrix} C_{\vec{k}, \sigma}^A \\ \vdots \\ C_{\vec{k}, \sigma}^A \\ C_{\vec{k}, \sigma}^B \\ \vdots \\ C_{\vec{k}, \sigma}^B \end{pmatrix}$

Al ser un hamiltoniano cuadrático escribimos:

$$\hat{H} = \frac{1}{2} \vec{d}^\dagger \vec{D} \vec{d} + \text{const.}$$

Por lo cual la matriz dinámica se puede escribir:

$$\vec{D} = \begin{pmatrix} E_{\vec{k}, \sigma}^0 \mathbb{I}_{N \times N} & \begin{pmatrix} \vec{\eta}_{\vec{k}, \sigma} \\ \vdots \end{pmatrix} & \mathbb{O}_{N \times N} & \begin{pmatrix} \vec{\eta}_{\vec{k}, \sigma}^\dagger \\ \vdots \end{pmatrix} \\ \begin{pmatrix} -\vec{\eta}_{\vec{k}, \sigma}^* \\ \vdots \end{pmatrix} & \hbar \omega_c & \begin{pmatrix} \vec{\eta}_{\vec{k}, \sigma} \\ \vdots \end{pmatrix} & 0 \\ \mathbb{O}_{N \times N} & \begin{pmatrix} \vec{\eta}_{\vec{k}, \sigma}^* \\ \vdots \end{pmatrix} & E_{\vec{k}, \sigma}^0 \mathbb{I}_{N \times N} & \begin{pmatrix} -\vec{\eta}_{\vec{k}, \sigma}^* \\ \vdots \end{pmatrix} \\ \begin{pmatrix} \vec{\eta}_{\vec{k}, \sigma}^* \\ \vdots \end{pmatrix} & 0 & \begin{pmatrix} -\vec{\eta}_{\vec{k}, \sigma} \\ \vdots \end{pmatrix} & \hbar \omega_c \end{pmatrix}$$

Labels in the diagram:

- Top-left: $E_{\vec{k}, \sigma}^0 \mathbb{I}_{N \times N}$
- Top-middle: $\begin{pmatrix} \vec{\eta}_{\vec{k}, \sigma} \\ \vdots \end{pmatrix}$ (labeled $\vec{\eta}_{\vec{k}, \sigma}$)
- Top-right: $\mathbb{O}_{N \times N}$
- Top-far-right: $\begin{pmatrix} \vec{\eta}_{\vec{k}, \sigma}^\dagger \\ \vdots \end{pmatrix}$ (labeled $\vec{\eta}_{\vec{k}, \sigma}^\dagger$)
- Middle-left: $\begin{pmatrix} -\vec{\eta}_{\vec{k}, \sigma}^* \\ \vdots \end{pmatrix}$ (labeled $-\vec{\eta}_{\vec{k}, \sigma}^*$)
- Middle: $\hbar \omega_c$
- Middle-right: $\begin{pmatrix} \vec{\eta}_{\vec{k}, \sigma} \\ \vdots \end{pmatrix}$ (labeled $\vec{\eta}_{\vec{k}, \sigma}$)
- Middle-far-right: 0
- Bottom-left: $\mathbb{O}_{N \times N}$
- Bottom-middle: $\begin{pmatrix} \vec{\eta}_{\vec{k}, \sigma}^* \\ \vdots \end{pmatrix}$ (labeled $\vec{\eta}_{\vec{k}, \sigma}^*$)
- Bottom-right: $E_{\vec{k}, \sigma}^0 \mathbb{I}_{N \times N}$
- Bottom-far-right: $\begin{pmatrix} -\vec{\eta}_{\vec{k}, \sigma}^* \\ \vdots \end{pmatrix}$ (labeled $-\vec{\eta}_{\vec{k}, \sigma}^*$)
- Far-bottom: $\begin{pmatrix} \vec{\eta}_{\vec{k}, \sigma}^* \\ \vdots \end{pmatrix}$ (labeled $\vec{\eta}_{\vec{k}, \sigma}^*$)
- Far-bottom-right: $\begin{pmatrix} -\vec{\eta}_{\vec{k}, \sigma} \\ \vdots \end{pmatrix}$ (labeled $-\vec{\eta}_{\vec{k}, \sigma}$)
- Far-bottom-far-right: $\hbar \omega_c$

Sin embargo para encontrar las energías debemos diagonalizar:

$$\hat{H} = \begin{pmatrix} E_{k0} \mathbb{1}_{N \times N} & \begin{matrix} \vec{\eta}_{k0} \\ \vdots \end{matrix} & \mathbb{O}_{N \times N} & \begin{matrix} \vec{\eta}_{k0} \\ \vdots \end{matrix} \\ \begin{matrix} \leftarrow -\vec{\eta}_{k0}^* \end{matrix} & \hbar \omega_c & \begin{matrix} \leftarrow \vec{\eta}_{k0} \end{matrix} & \mathbb{O} \\ \mathbb{O}_{N \times N} & \begin{matrix} \vec{\eta}_{k0}^* \\ \vdots \end{matrix} & -E_{k0} \mathbb{1}_{N \times N} & \begin{matrix} \vec{\eta}_{k0}^* \\ \vdots \end{matrix} \\ \begin{matrix} \leftarrow -\vec{\eta}_{k0}^* \end{matrix} & \mathbb{O} & \begin{matrix} \leftarrow \vec{\eta}_{k0} \end{matrix} & \hbar \omega_c \end{pmatrix}$$

Sea $\vec{V} = \begin{pmatrix} \vec{\mu} \\ \vec{\nu} \\ \varphi \end{pmatrix} + \mu, \nu \in \mathbb{C}^{2N+1}$, entonces:

Indexamos: $\vec{\mu} = \begin{pmatrix} \mu_{k1A} \\ \mu_{k2A} \\ \vdots \\ \mu_{kNB} \\ \mu_{\phi} \end{pmatrix}, \vec{\nu} = \begin{pmatrix} \nu_{k1A} \\ \nu_{k2A} \\ \vdots \\ \nu_{kNB} \\ \nu_{\phi} \end{pmatrix}$

Por lo que:

$$\hat{H} \vec{V} = \hbar \omega \vec{V}$$

Está dado por las ecuaciones:

$$\text{Bloques } \left\{ \begin{array}{l} \mu_{k\sigma} (E_{k\sigma} - \omega) - \gamma_{k\sigma} \mu \phi + \gamma_{k\sigma} \nu \phi = 0 \\ \gamma_{k\sigma} (-E_{k\sigma} - \omega) - \gamma_{k\sigma}^* \mu \phi + \gamma_{k\sigma}^* \nu \phi = 0 \end{array} \right.$$

$$\left\{ \begin{array}{l} \hbar(\omega_c - \omega) \mu \phi - \sum_{k\sigma} \gamma_{k\sigma}^* \mu_{k\sigma} + \sum_{k\sigma} \gamma_{k\sigma} \nu_{k\sigma} = 0 \\ \hbar(-\omega_c - \omega) \nu \phi - \sum_{k\sigma} \gamma_{k\sigma}^* \mu_{k\sigma} + \sum_{k\sigma} \gamma_{k\sigma} \nu_{k\sigma} = 0 \end{array} \right.$$

$$\Rightarrow (\omega_c - \omega) \mu \phi = -(\omega_c + \omega) \nu \phi$$

$$\text{2. } \nu \phi = \frac{(\omega - \omega_c)}{\omega + \omega_c} \mu \phi$$

Reemplazando en las expresiones anteriores:

$$\left\{ \begin{array}{l} \mu_{k\sigma} (E_{k\sigma} - \omega) - \gamma_{k\sigma} \mu \phi \left(\frac{2\omega_c}{\omega + \omega_c} \right) = 0 \\ \gamma_{k\sigma} (-E_{k\sigma} - \omega) - \gamma_{k\sigma}^* \mu \phi \left(\frac{2\omega_c}{\omega + \omega_c} \right) = 0 \end{array} \right.$$

$$\left\{ \begin{array}{l} \mu_{k\sigma} = \frac{2 \gamma_{k\sigma} \omega_c}{(E_{k\sigma} - \omega)(\omega + \omega_c)} \mu \phi \\ \gamma_{k\sigma} = \frac{-2 \gamma_{k\sigma}^* \omega_c}{(E_{k\sigma} + \omega)(\omega + \omega_c)} \mu \phi \end{array} \right.$$

Por tanto:

$$\mu \phi \left((\omega_c - \omega) - \sum_{k\sigma} \frac{2 |\gamma_{k\sigma}|^2 \omega_c}{(E_{k\sigma} - \omega)(\omega + \omega_c)} - \sum_{k\sigma} \frac{2 |\gamma_{k\sigma}|^2 \omega_c}{(E_{k\sigma} + \omega)(\omega + \omega_c)} \right) = 0$$

Buscamos soluciones no nulas, portanto:

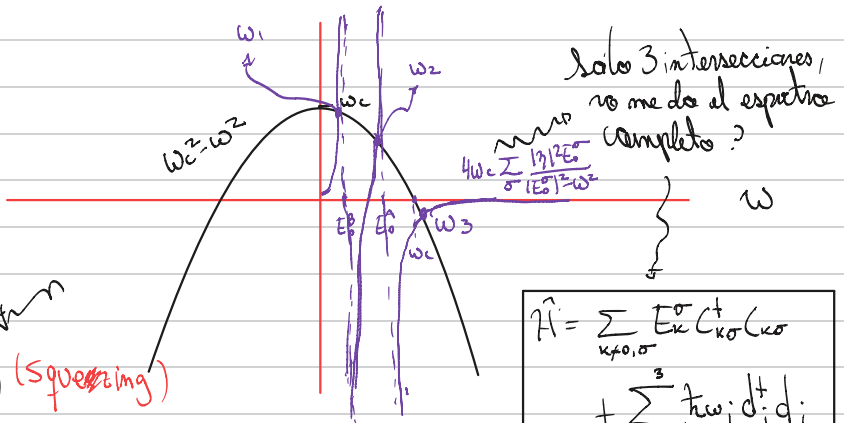
$$\omega_c^2 - \omega^2 = 4\omega_c \sum_{\mathbf{k}, \sigma} \frac{|\eta_{\mathbf{k}\sigma}|^2 E_{\mathbf{k}\sigma}}{|\epsilon_{\mathbf{k}\sigma}|^2 - \omega^2}$$

Es una ecuación implícita para las autoenergías del problema.

• Ejemplo

Supongamos $\eta_{\mathbf{k}\sigma} = \eta \delta_{\mathbf{k},0}$ ($\eta_A^0 = \eta_B^0$)

$$\omega_c^2 - \omega^2 = 4\omega_c \sum_{\sigma} \frac{|\eta|^2 E_{\sigma}^0}{|\epsilon_{\sigma}^0|^2 - \omega^2}$$



Energía del fotón aumenta, (squeezing) la de los modos básicos disminuye.

Las modas ω_1 y ω_2 corresponden a los LP y UP en $\mathbf{k}=0$ y ω_3 al fotón vestido por la interacción.

$$\hat{H} = \sum_{\mathbf{k} \neq 0, \sigma} E_{\mathbf{k}\sigma}^0 c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} + \sum_{j=1}^3 \hbar \omega_j d_j^\dagger d_j$$